THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 11: **Keeping Time with the Skies** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The Earth's axis is tilted by about: 1

(a) 0 degrees **(b)** 90 degrees **(c)** 45 degrees **(d)** 23.5 degrees

2. The direction of the shadow of a stick in the morning is towards the: 1

(a) south only **(b)** east **(c)** north only **(d)** west

3. The period from new moon to full moon is about: 1

(a) two weeks **(b)** one day **(c)** one hour **(d)** one year

4. The longest day in the northern hemisphere occurs on about: 1

(a) 21 December **(b)** 1 January **(c)** 21 June **(d)** 21 March

5. The Pole Star (Polaris) appears to: 1

(a) set in the west **(b)** stay almost fixed in the northern sky
(c) rise in the east **(d)** move fastest

6. Bright half of the Moon is always facing the: 1

(a) Pole Star **(b)** Sun **(c)** Mars **(d)** Earth

7. The day and night are almost equal on: 1

(a) 1 May **(b)** 21 March and 23 September
(c) 21 June **(d)** 21 December

8. The stars appear to move across the night sky because of the: 1

(a) revolution of Jupiter **(b)** movement of the Moon
(c) rotation of the Earth **(d)** motion of the stars around the Earth

9. The Moon is waxing when its lit part is: 1

(a) decreasing **(b)** zero **(c)** increasing **(d)** constant

10. The shortest day in the northern hemisphere is on about: 1

(a) 21 June **(b)** 23 September **(c)** 21 March **(d)** 22 December

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the number of days in a lunar month (approximately). 2

12. What is a gnomon? Where is it used? 2

13. Why do we always see the same face of the Moon? 2

14. What are the solstices and equinoxes? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the Moon seems to change shape in the sky. 3

16. Explain why the length of day and night changes through the year. 3

17. State three differences between day length at the equator and at higher latitudes during summer. 3

OR

Differentiate between a solar and a lunar calendar (three points). 3

18. State three uses of calendars. 3

OR

Explain how ancient people used the sky to keep time and track seasons. Give two examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the changes in the length and direction of the shadow of a vertical stick during the day and in different seasons. 5

OR

Describe how the stars and constellations change with seasons, and name three constellations. 5

20. Explain the solstices and equinoxes with dates and the position of the Sun and length of day. 5

OR

Explain how an eclipse of the Sun or Moon occurs and why not every month. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND